Assessment Worksheet

DBMS Lab (BCSC0802)

We are designing a **College Database** with the following entities:

1. **Department**
2. **Student**
3. **Course**

Student(student_id NUMBER PRIMARY KEY, student_name VARCHAR2(30) NOT NULL, email VARCHAR2(40) UNIQUE, age NUMBER , dept_id NUMBER FOREIGN KEY REFERENCES Department(dept_id), admission_date DATE DEFAULT SYSDATE)

Sample data:

INSERT INTO Student VALUES (101, 'Amit Kumar', 'amit@gmail.com', 20, 1, SYSDATE);
INSERT INTO Student VALUES (102, 'Neha Sharma', 'neha@gmail.com', 22, 2, SYSDATE);
INSERT INTO Student (student_id, student_name, age, dept_id) VALUES (103, 'Rahul Verma', 19, 1);

Department (dept_id NUMBER PRIMARY KEY, dept_name VARCHAR2(30) NOT NULL UNIQUE, location VARCHAR2(20) DEFAULT 'Main Block')

Sample data:

INSERT INTO Department VALUES (1, 'Computer Science', 'Block A');

INSERT INTO Department VALUES (2, 'Electronics', 'Block B');

INSERT INTO Department (dept_id, dept_name) VALUES (3, 'Mechanical');

Course(course_id NUMBER PRIMARY KEY, course_name VARCHAR2(40) NOT NULL, credits NUMBER CHECK (credits BETWEEN 1 AND 5), dept_id NUMBER NOT NULL FOREIGN KEY REFERENCES Department(dept_id))

Sample data:

INSERT INTO Course VALUES (201, 'DBMS', 4, 1);

INSERT INTO Course VALUES (202, 'Data Structures', 5, 1);

INSERT INTO Course VALUES (203, 'Digital Electronics', 3, 2);

A. Design the tables based on the schema description, insert sample data
B. Solve the following queries:

1. Display all records of the **Student** table.
2. Display the **student name, email, and age** of all students.
3. Display details of students whose **age is greater than 20**.
4. Display students who belong to a **specific department**.
5. Display students whose **name starts with the letter 'A'**.
6. Display students belonging to **department 1 or department 3**.
7. Display students whose **age is between 18 and 22**.
8. Display all students **sorted in ascending order of student name**.
9. Display **distinct department IDs** from the Student table.
10. Display **student names along with their department names**.
11. Add a **phone number column** to the Student table.
12. Modify the **size of the student name column** in the Student table.
13. Delete records of students who **belong to a particular department**.
14. Add a **CHECK constraint** to ensure student age is not less than 18.
15. Drop the **foreign key constraint** from the Student table.